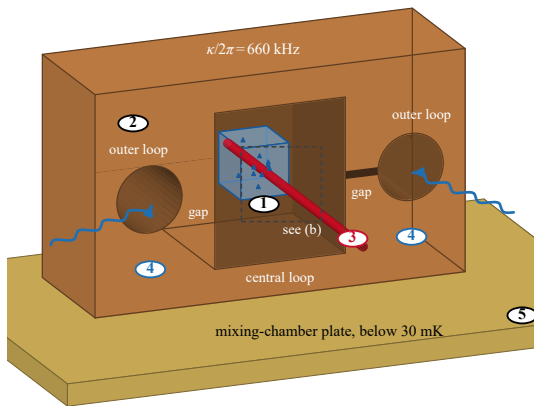
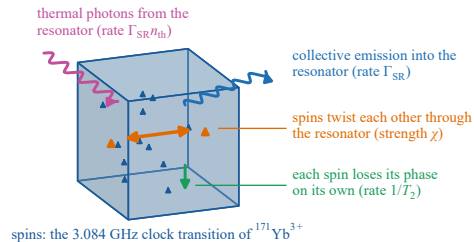


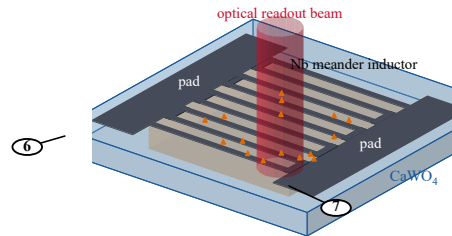
**(a)** the demonstrated experiment: crystal in a loop-gap microwave resonator



**(b)** what the model follows inside the crystal



**(c)** proposed planar superconducting resonator on the same crystal (not built)



**Key**

- ①  $^{171}\text{Yb}^{3+}:\text{CaWO}_4$  crystal,  $4.4 \times 4.6 \times 5$  mm, 4.96 ppm  $^{171}\text{Yb}$ , glued in the central loop
- ② metal loop-gap resonator: central loop, two outer loops, narrow gaps ( $g/2\pi = 15$  mHz,  $V_m = 275$  mm<sup>3</sup>)
- ③ 973 nm light through the crystal: optical pumping into  $|\downarrow\rangle$  and absorption readout of the populations
- ④ microwave antennas at the outer loops: spin control pulses and transmission measurement
- ⑤ mixing-chamber plate of the dilution refrigerator (spin temperature 80 mK,  $n_{\text{th}} = 0.19$ )
- ⑥  $\text{CaWO}_4$  chip with a niobium film patterned into a meander inductor between two capacitor pads
- ⑦ read-out volume under the inductor, where the microwave field is nearly uniform (coupling spread  $D$ )